20 In London, 1 litre of petrol costs 108.9p
In New York, 1 US gallon of petrol costs $2.83

1 US gallon = 3.785 litres
£1 = $1.46

In which city is petrol better value for money, London or New York?
You must show your working.

(Total for Question 20 is 3 marks)

21 A gold bar has a mass of 12.5 kg.

The density of gold is 19.3 g/cm^3

Work out the volume of the gold bar.
Give your answer correct to 3 significant figures.

12.5 kg = 12500 g. In the table, cm3 means cm^3.

g	cm3
19.3	1
12500	$\frac{125000}{193}$

$\times \frac{125000}{193}$ $\times \frac{125000}{193}$

Volume = mass ÷ density

= 12500 ÷ 19.3

= 647.668393... cm^3

2 marks total
Correct units and division

1 mark
648 cm^3 to 3 significant figures

648 cm^3

(Total for Question 21 is 3 marks)

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